Computational Mechanics:

3. Prediction

ILIAD INTENSIVE – SPRING 2026

Outline:

1) Belief state

2) Next-token (probability) vector

3) Mixed state presentation

4) GHMM prediction

Belief state

How should we predict tokens emitted from an HMM?

↪ By defⁿ the next-token distribution is a function of the current distribution over hidden states.

↪ Use this to make predictions

Bayes' rule tells us that the posterior prob. of occupying state s_j after observing x is:

$$P(s_j \mid x) = \frac{P(x, s_j)}{P(x)} = \frac{\sum_{s_0 \in S} P(s_0)\, P(x, s_j \mid s_0)}{P(x)}$$

hidden state

observed tokens

Similarly, posterior prob. of occupying s_j after observing the sequence $W = w_1 w_2 \cdots w_n$ is:

$$P(s_j | W) = \frac{\sum_{s^0, \ldots, s^{n-1}} P(s_0) P(w_1, s^1 | s^0) P(w_2, s^2 | s^1) \cdots P(w_n, s_j | s^{n-1})}{P(W)}$$

Expressing this object in terms of HMM data:

$$P(s_j | W) = \left(\frac{\eta^{(\phi)} T^{(w_1)} T^{(w_2)} \cdots T^{(w_n)}}{\underbrace{\eta^{(\phi)} T^{(w_1)} T^{(w_2)} \cdots T^{(w_n)} \mathbf{1}}_{\text{row-vector}}} \right)_j$$

where we order the hidden states $S = \{s_0, s_1, \ldots\}$.

This motivates the def$^{\underset{\sim}{n}}$ of the <u>belief state</u>:

$$\eta^{(w)} := \frac{\eta^{(\phi)} T^{(w_1)} T^{(w_2)} \cdots T^{(w_n)}}{\eta^{(\phi)} T^{(w_1)} T^{(w_2)} \cdots T^{(w_n)} \mathbb{1}}, \quad w = w_1 w_2 \cdots w_n \in \mathcal{X}^*$$

whose $j^{\underline{th}}$ element corresponds to the prob. that the hidden state is s_j after obs. w.

Suppose we observe a further token $x \in \mathcal{X}$, the predictive vector update rule is simple:

$$\eta^{(w)} \longmapsto \eta^{(wx)} = \frac{\eta^{(w)} T^{(x)}}{\eta^{(w)} T^{(x)} \mathbb{1}}$$

Next-token (probability) vector

For predicting next-token emissions, we actually care about:

$$P(x_j \mid w) = \frac{P(w x_j)}{P(w)} = \frac{\eta^{(\phi)} T^{(w_1)} T^{(w_2)} \cdots T^{(w_n)} T^{(x_j)} \mathbb{1}}{\eta^{(\phi)} T^{(w_1)} T^{(w_2)} \cdots T^{(w_n)} \mathbb{1}}$$

next-token

observed tokens

This motivates the def$^{\underline{n}}$ of the next-token vector:

$$p^{(w)} := \left(\frac{\eta^{(\phi)} T^{(w_1)} T^{(w_2)} \cdots T^{(w_n)} T^{(x_j)} \mathbb{1}}{\eta^{(\phi)} T^{(w_1)} T^{(w_2)} \cdots T^{(w_n)} \mathbb{1}} \right)_{j=0,1,\ldots}$$

where we order the tokens $\mathcal{X} = \{x_0, x_1, \ldots\}$.

The j^{th} element of $p^{(w)}$ corresponds to the prob. that the next-token is x_j after obs. w.

The belief state appears directly in the def$^{\underset{\sim}{n}}$ of the next-token vector:

$$p_j^{(w)} := \frac{\eta^{(\phi)} T^{(w_1)} T^{(w_2)} \ldots T^{(w_n)} T^{(x_j)} 1}{\eta^{(\phi)} T^{(w_1)} T^{(w_2)} \ldots T^{(w_n)} 1} = \eta^{(w)} T^{(x_j)} 1$$

It follows that there exists matrix that maps $\eta^{(w)} \mapsto p^{(w)}$ i.e.:

$$p^{(w)} = \eta^{(w)} A$$

where the matrix elements of A are:

$$A_{ij} := \sum_{k=0}^{|S|-1} T_{ik}^{(x_j)}, \qquad \begin{matrix} i = 0, 1, \ldots |S|-1 \\ j = 0, 1, \ldots |\mathcal{X}|-1 \end{matrix}.$$

$$\mu^{(w)} \xmapsto{A} \rho^{(w)}$$

Hidden dynamic

Observations

If A is <u>injective</u>:

If A is <u>not injective</u>:

Belief states are also useful for all future preds.

Let $w^{(p)} = w_1^{(p)} w_2^{(p)} \dots w_m^{(p)}$ & $w^{(f)} = w_1^{(f)} w_2^{(f)} \dots w_n^{(f)}$:

$$P(w^{(f)} \mid w^{(p)}) = \frac{\eta^{(\varnothing)} T^{(w_1^{(p)})} \cdots T^{(w_m^{(p)})} T^{(w_1^{(f)})} \cdots T^{(w_n^{(f)})} \mathbb{1}}{\eta^{(\varnothing)} T^{(w_1^{(p)})} \cdots T^{(w_m^{(p)})} \mathbb{1}}$$

$$= \eta^{(w^{(p)})} T^{(w_1^{(f)})} \cdots T^{(w_n^{(f)})} \mathbb{1}$$

As in the next-token case there exists a linear map from $\eta^{(w^{(p)})}$ to all future predictions.

Mixed state presentation

For any HMM, the deterministic update rule $\eta^{(w)} \mapsto \eta^{(wx)}$ defines a meta-dynamic over beliefs called the mixed-state presentation (MSP).

The MSP is itself a (unifilar) HMM:

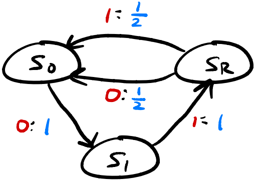
$$\eta^{(w)} \xrightarrow{x \,:\, p} \eta^{(wx)} \qquad p = P(x \mid w)$$

whose hidden states are the belief states.

[For unifilar HMMs, an ε-machine is the MSP of the corresponding minimal HMM.]

Example 1: (zero-one-random $p=\frac{1}{2}$)

Recall:

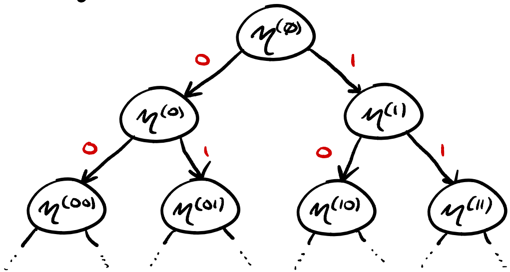


$$T^{(0)} = \begin{array}{c|l} \begin{matrix} S_0 & S_1 & S_R \end{matrix} & \\ \hline \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ \frac{1}{2} & 0 & 0 \end{bmatrix} & \begin{matrix} S_0 \\ S_1 \\ S_R \end{matrix} \end{array}$$

$$T^{(1)} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ \frac{1}{2} & 0 & 0 \end{bmatrix}$$

$$\boxed{\begin{gathered} \eta^{(w)} = \frac{\eta^{(\emptyset)} T^{(w)}}{\eta^{(\emptyset)} T^{(w)} \mathbb{1}} \\ T^{(w)} := T^{(w_1)} \cdots T^{(w_n)} \end{gathered}}$$

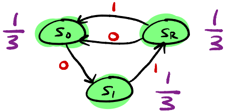
Deriving the MSP compute:



$\rightsquigarrow$ Identify nodes where

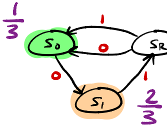
$$\eta^{(w)} = \eta^{(w')}$$

Fix: $\mathcal{M}^{(\phi)} = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{bmatrix}$

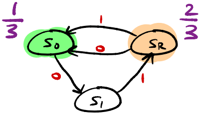


and compute:

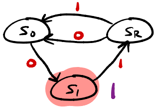
$\mathcal{M}^{(0)} = \begin{bmatrix} \frac{1}{3} & \frac{2}{3} & 0 \end{bmatrix}$



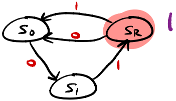
$\mathcal{M}^{(1)} = \begin{bmatrix} \frac{1}{3} & 0 & \frac{2}{3} \end{bmatrix}$



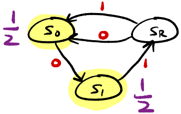
$\mathcal{M}^{(00)} = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix}$



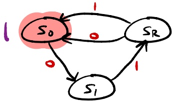
$\mathcal{M}^{(01)} = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}$



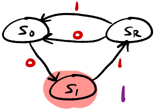
$\mathcal{M}^{(10)} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix}$



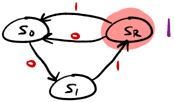
$\mathcal{M}^{(11)} = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$



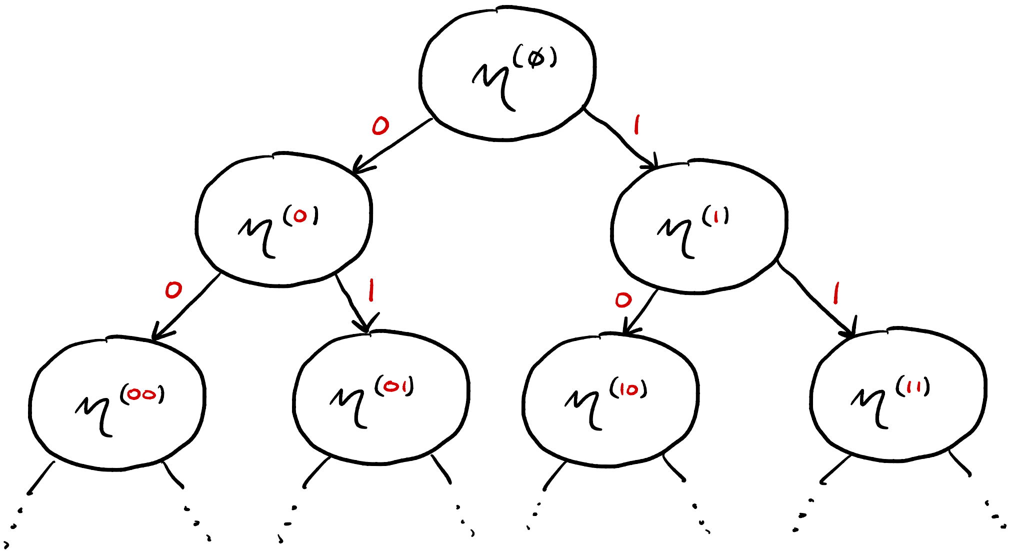
$\mathcal{M}^{(100)} = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix}$



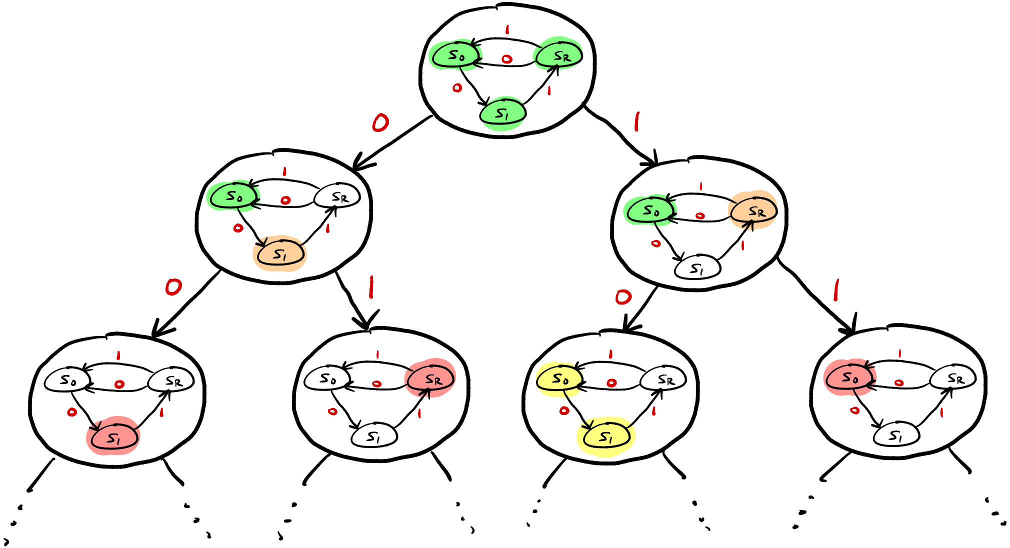
$\mathcal{M}^{(101)} = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}$



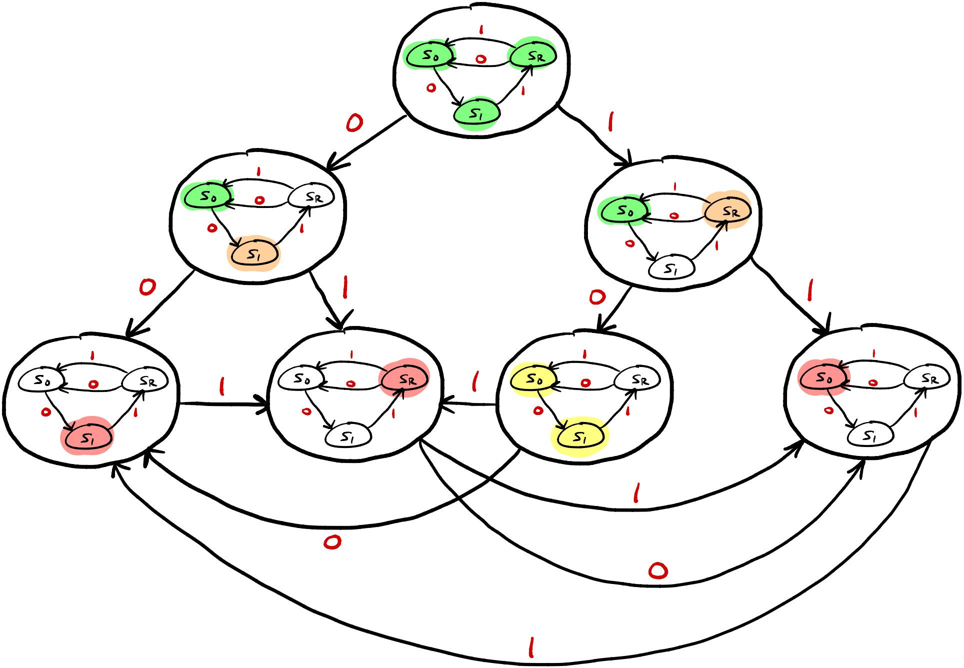
Inference processes converges!

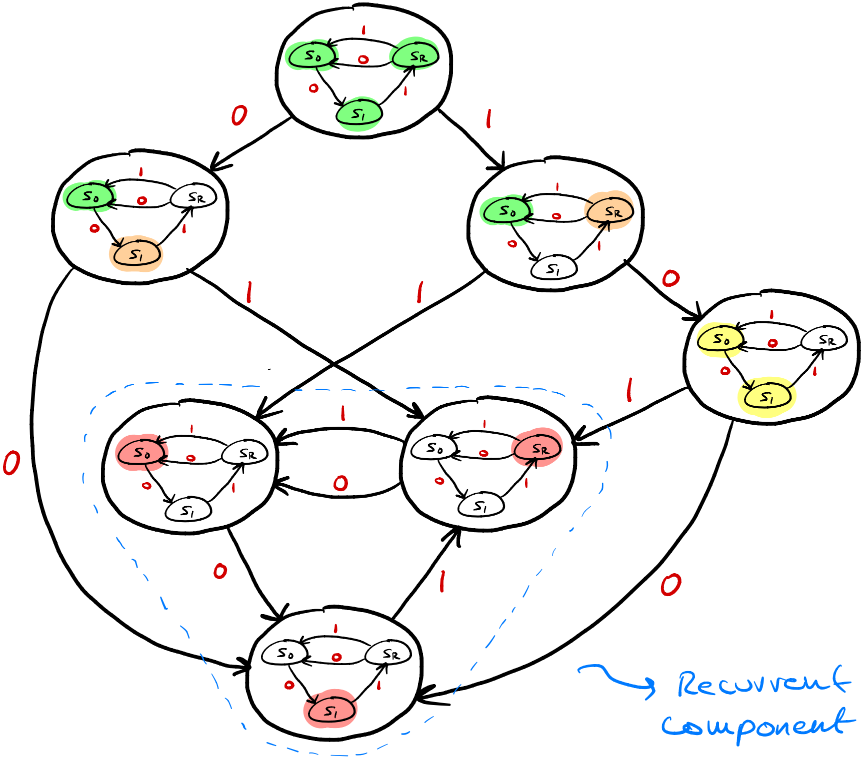


$\eta^{(\emptyset)}$
0
1
$\eta^{(0)}$
$\eta^{(1)}$
0
1
0
1
$\eta^{(00)}$
$\eta^{(01)}$
$\eta^{(10)}$
$\eta^{(11)}$

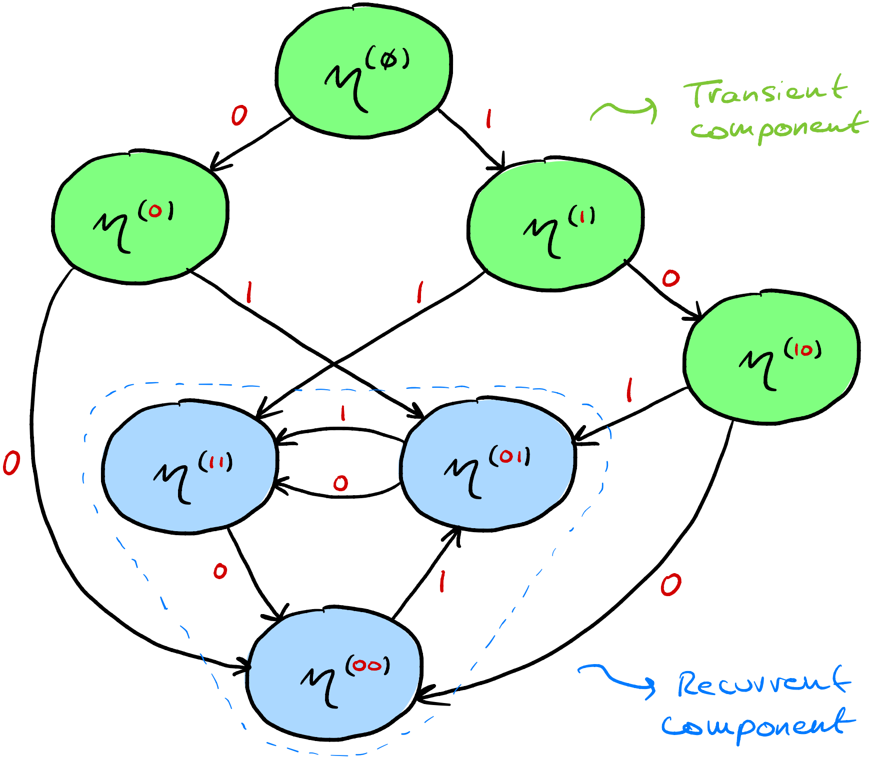


0
1
S_0
S_R
S_1





S_0
S_R
S_1
0
1
Recurrent component

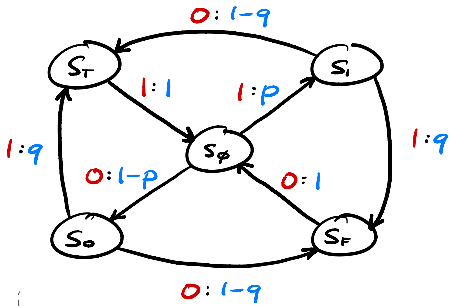


Transient component
Recurrent component

Example 2: (random–random–XOR)

$$S = \{s_\phi, s_0, s_1, s_F, s_T\}$$

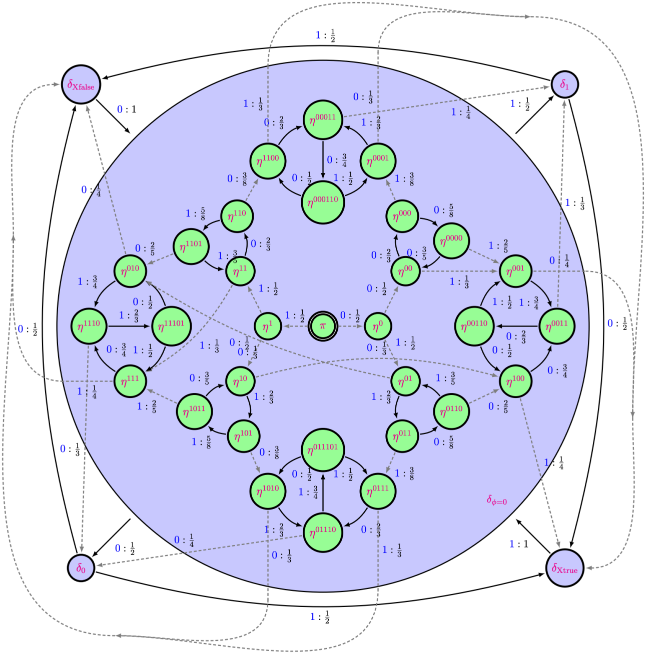
$$\mathcal{X} = \{0, 1\}$$



$$T^{(0)} = \begin{array}{cc} \begin{array}{ccccc} s_\phi & s_0 & s_1 & s_F & s_T \end{array} & \\ \begin{bmatrix} 0 & 1-p & 0 & 0 & 0 \\ 0 & 0 & 0 & 1-q & 0 \\ 0 & 0 & 0 & 0 & 1-q \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} & \begin{array}{c} s_\phi \\ s_0 \\ s_1 \\ s_F \\ s_T \end{array} \end{array}$$

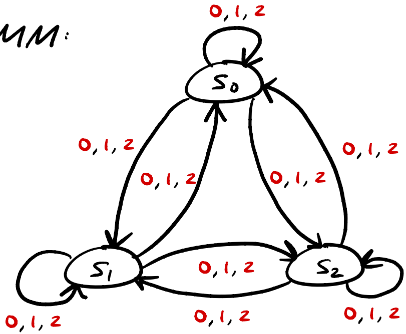
$$T^{(1)} = \begin{bmatrix} 0 & 0 & p & 0 & 0 \\ 0 & 0 & 0 & 0 & q \\ 0 & 0 & 0 & q & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix}$$

The corresponding MSP:



Example 3: (Mess3)

HMM:

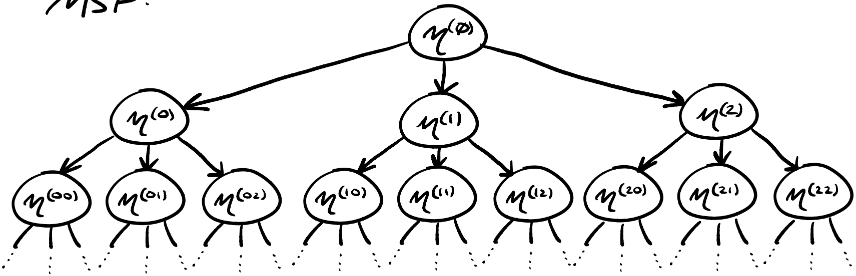


$$T^{(0)} = \begin{array}{c|c} \begin{array}{ccc} S_0 & S_1 & S_2 \end{array} & \\ \begin{bmatrix} \alpha y & \beta x & \beta x \\ \alpha x & \beta y & \beta x \\ \alpha x & \beta x & \beta y \end{bmatrix} & \begin{array}{c} S_0 \\ S_1 \\ S_2 \end{array} \end{array}$$

$$T^{(1)} = \begin{bmatrix} \beta y & \alpha x & \beta x \\ \beta x & \alpha y & \beta x \\ \beta x & \alpha x & \beta y \end{bmatrix}$$

$$T^{(1)} = \begin{bmatrix} \beta y & \beta x & \alpha x \\ \beta x & \beta y & \alpha x \\ \beta x & \beta x & \alpha y \end{bmatrix}$$

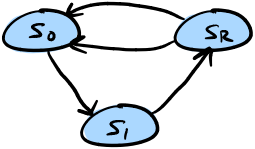
MSP:



Inference does not converge!

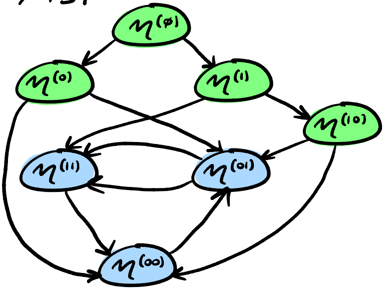
Summary:

HMM:



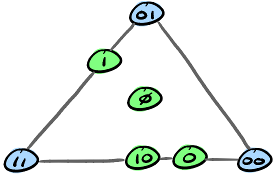
inference $\longrightarrow$

MSP:



geometry $\swarrow$

Belief-state geometry:



linear map $\longrightarrow$

Next-token geometry:

GHMM prediction:

Throughout we have considered prediction of HMM data. Fortunately, all of the relevant objects have straightforward analogues in the generalised setting.

Recall that a GHMM consists of:

$(T^{(x)})_{x\in\mathcal{X}}$ $\quad\eta^{(\phi)}$ – initial state $\quad\varphi$ – final state

Belief state (aka predictive vector):

$$\eta^{(w)} := \frac{\eta^{(\phi)}T^{(w_1)}T^{(w_2)}\cdots T^{(w_n)}}{\eta^{(\phi)}T^{(w_1)}T^{(w_2)}\cdots T^{(w_n)}\varphi}, \quad w = w_1 w_2 \cdots w_n \in \mathcal{X}^*$$

Next-token vector:

$$p^{(w)} = \left(\eta^{(w)} T^{(x_j)} \varphi\right)_{j=0,1,\dots}$$

Linear map $\eta^{(w)} \xmapsto{A} p^{(w)}$:

$$A_{ij} := \sum_{k=0}^{d-1} T_{ik}^{(x_j)} \varphi_k \qquad \begin{aligned} i &= 0, 1, \dots, d-1 \\ j &= 0, 1, \dots, |\mathcal{X}|-1 . \end{aligned}$$

A GHMM does not have 'underlying' hidden states so $\eta^{(w)}$ is not stochastic and thus cannot be viewed as a posterior distribution.

↪ Despite this the belief state remains similarly instrumental at making token predictions.

Questions?